
WANDERERS **TEST PLAN**

Version <1.3>

<01/04/2025>

VERSION HISTORY

Version #	Implemented By	Revision Date	Approved By	Approval Date	Reason
1.0	Raghav	30/03/2025	Isaac	01/04/25	Initial Version
1.1	Miko	31/3/2025	Isaac	01/04/25	Rephrased and edited item pass/fail section
1.2	Seo Jin	01/4/2025	Isaac	01/04/25	Updated Approach, Suspension Criteria, Staff needs, and responsibilities.
1.3	Isaac	01/4/2025	Seo Jin	01/04/25	Edited parts on schedule and testing stratgy

UP Template Version: 01/04/25

1 - Test Plan Identifier

The unique identifier for this master test plan of the Wanderers web application project is **WAND-MTP-1.0**, where:

WAND: Refers to Wanderers project

MTP: Refers to Master Test Plan

1.0: Indicates the version number of the document

Throughout the testing process, we will be applying the test documentation specifications described in the IEEE Standard 829-2008 for software test documentation. The plan contains information about the resources and procedures of testing of **Wanderers**, and how the testing process is controlled, and configuration managed throughout the product lifecycle.

2 – Introduction

This document serves as the **Master Test Plan** for the Wanderers project. It provides a high-level overview of the testing approach, scope, and objectives for the system. The purpose of this plan is to define the testing approach, ensuring that the system meets functional and non-functional requirements before deployment. This ensures that the application provides a functional and user-friendly experience without any inconveniences and bugs.

This document serves as a high-level guide for all testing activities, including **unit testing**, **integration testing**, **system testing**, and **user acceptance testing (UAT)**.

1. **Unit Testing:** Test cases created by the Quality Management team will be used to test each individual unit.
2. **Integration Testing:** Units which have been tested will be tested in a combined manner to expose faults in integration.
3. **System Testing:** Independent external testers will be asked to test the entire system as a whole and their feedback will be taken.
4. **User Acceptance Testing:** Prospective end-users will be asked to use the application, and their feedback will be taken.

The above-mentioned testing activities are done using the **white box** and **black box** approach.

For white box testing, we will be using the control flow testing to perform the test cases. The test team will follow the test cases as closely as possible to ensure that any bugs or problems are to be highlighted and fixed.

For black box testing, a certain number of end users who do not have any prior knowledge of Wanderers will test out the beta version of the application and we will obtain feedback from them. Feedback will include but not limit to product quality, user friendliness and bugs.

3 – Test Items (Functions)

The Wanderers web application will undergo comprehensive testing to ensure its functionality, performance, and reliability. The following core components of the system will be tested as part of this master test plan:

1. User Authentication and Account Management

- User registration, login and logout functionalities
- Password reset and account recovery process

2. Trip planning and collaboration

- Creation and management of trip itineraries
- Invitation and user participation in trips

3. Expense Tracking and Budgeting

- Adding, editing and deleting expenses within trip

4. Communication and Notifications

- In-app messaging and group chat functionality

5. User Interface and Navigation

- Responsive design and usability across devices
- Proper functioning of menu, buttons and input fields

6. Data Management and Synchronization

- Proper storage and retrieval of trip-related data
- Synchronization across multiple devices and users
- Data integrity and consistency checks

7. Performance & Scalability

- Load testing for concurrent users accessing trip data

- Response time for key actions (e.g., adding an expense, updating itinerary)
- System performance under heavy traffic

8. Security & Compliance

- Protection of user data and secure handling of sensitive information
- Encryption of stored and transmitted data
- Compliance with relevant privacy laws and regulations

These higher-level test items are aligned with the current latest software version of Wanderers and will be updated based on future releases. Any necessary configuration requirements and dependencies will be documented as part of the test execution phase.

A detailed listing of programs, databases, screens and reports will be provided in the system and detailed design documents.

4 – Software Risk Issues

There are several software risk issues recognized which can have a direct impact on the web application and need to be handled carefully. The issues are detailed below.

1. Third-Party Components

- Assess risks associated with integrating third-party applications or APIs for specific features of the application
- Google Places API

2. Database Security and Safety

3. Backup and Recovery of Files

4. Reliability of web hosting service

5 – Features to be tested

Features will be rated based on their risk into three levels: High, Medium, and Low. High level risks have the highest importance and must be tested and debugged as soon as possible. Medium level features are to be tested only after high level bugs. Low level features are not as important and will only be tested once all other features are working properly

No.	Features	Risk Level (Low, Medium, High)
1	Register Account	High
2	Registered User Login	High
3	Registered User Logout	Medium
4	Reset User Account Password	Low
5	Create Itinerary	High
6	View Itinerary Details	Low
7	Edit Itinerary Details	High
8	Delete Itinerary	Medium
9	Create Activity	High
10	View Activity Details	Low
11	Edit Activity Details	High
12	Delete Activity	Medium

6 – Features not to be tested

The following features will not be explicitly tested. All testing in these areas will be done indirectly as a result of other testing efforts.

- Third Party APIs
 - Google Places API is called for features such as “View Itinerary Details” and “View Activity Details”. These features will only be tested for integration, and not for Google API’s functionality.
- Stable Features
 - Features that have not changed from previous versions and have been validated in prior testing cycles will not be re-tested.

7 – Approach (Strategy)

The testing for Wanderers project consists of four primary level: Unit Testing, System Testing, Integration Testing, User Acceptance Testing

Unit testing will be conducted by developers as each module is completed. Automated unit tests will be written to validate individual components and functions. To ensure continuous validation, these tests will run each time a new commit is made via the CI/CD pipeline. Each

unit test must pass successfully before the module can be considered complete. Any defects identified will be documented and shared with the development team for resolution.

System testing will be conducted by the QA team in a controlled environment. This testing evaluates the entire application as a unified system, to ensure all components work together seamlessly. The testing will include both functional and non-functional tests to validate core features, and the test cases will be based on the functional requirements outlined in the SRS document.

Integration testing will be performed after unit testing to verify the interactions between integrated components. The primary focus will be on data flow between the frontend, backend, and external APIs. The QA team will carry out integration testing to detect issues related to component communication and interface mismatches. This phase will begin only after critical defects from unit testing have been addressed.

User Acceptance Testing will be conducted by selected end users under the supervision of the QA team. This phase will be taking place after system testing and will focus on real-world scenarios to validate usability and performance. This is to ensure that the application meets user expectations and functional requirements. Any feedback or issues identified during this testing will be documented and addressed by the team before the final release.

8 - Item Pass/Fail Criteria

The pass/fail criteria for the Wanderers Master Test Plan define the conditions under which the system is considered sufficiently tested and ready for release. These criteria ensure that all necessary testing levels are completed, defects are addressed, and the software meets functional and non-functional requirements.

For each test case, the proper input and the expected output is given to the tester, which is detailed in the **“Test Cases and Coverage” document**. The test case is only deemed to have passed if the result matches **exactly** with the expected output defined in the test case. If it is unclear whether the actual output matches the expected output, the QA manager will be consulted, who will consult the Project Manager as and when necessary.

The **completion criteria** for this plan are as follows:

1. All lower-level test plans (unit, integration, system, and acceptance testing) must be executed.
2. 100% of the unit test cases are completed
3. 95% of the integration testing cases are completed and 5% cases have minor defects

4. System Testing is done with 3 external testers
5. A minimum of 100 external users have given their feedback for User Acceptance Testing

9 - Suspension Criteria and Resumption Requirements

A. Critical System Failures

Testing will be suspended if there is a critical system failure that prevents any form of testing, such as database crashes, loss of network connectivity, or severe application crashes. Testing will resume once system stability is restored and verified. A root cause analysis will be conducted to ensure the issue does not recur, and corrective actions will be implemented and reviewed before proceeding.

B. Major Security Breach

Testing will be suspended if a significant security breach occurs during testing that could potentially compromise test data or the integrity of the test results. Testing will only resume after fully containing the breaches, addressing the vulnerabilities, and testing security enhancements. Consequently, a security audit will be conducted to validate the effectiveness of the corrective measures.

C. Unavailability of Critical Testing Tools

Testing will be suspended if essential testing tools or environments become unavailable for an extended period (e.g., due to licensing issues or hardware failures). Testing will only resume once the necessary tools or environments are fully operational. The project timeline will be reassessed to accommodate the delay.

D. High Defect Rates

Testing will be suspended if the defect rate exceeds a predetermined threshold that makes further testing impractical, such as when multiple critical bugs block core functionalities. Testing will resume after a significant number of critical and major defects have been fixed, and a stable build has been verified.

E. Regulatory Compliance Issues

Testing will be suspended if it is discovered that the application does not comply with critical regulatory requirements during testing. Testing will resume after necessary adjustments are made to ensure compliance, and approval from the compliance team is obtained.

F. Resource Unavailability

Testing will be suspended if there is a significant unavailability of key personnel or critical resources (e.g., third-party service downtime). Testing will resume once staffing levels are restored or required resources become available. If there are extensive delays, additional resources may be allocated to maintain the project schedule.

To promptly identify potential triggers for each of these criteria, thorough monitoring will be carried out, and all stakeholders will receive regular updates to stay informed.

10 – Test Deliverables

Test Deliverables are the test artifacts which are given to the stakeholders of a software project during the SDLC (Software Development Life Cycle). Some of the deliverables are provided before the testing phase commences and some are provided during the testing phase and rest after the testing phase is completed.

1. Test Cases and Test Coverage Report

- Test cases are the set of positive and negative executable steps of a test scenario which has a set of pre-conditions, test data, expected result, post-conditions, and actual results. This also contains the test results and the summary of test execution activities.

2. Test Plan Document

- Test plan document is a document which contains the plan for all the testing activities to be done to deliver a quality product. The test Plan document is derived from the Product Description, SRS, or Use Case documents for all future activities of the project. It is usually prepared by the Test Lead or Test Manager

3. Revision Logs

- The revision history of all the documents will be duly documented and stored

4. Defects log with implemented solutions

- All the defects identified will be documented and logged in the appropriate documents as well as the solution implemented for the respective defect.

11 – Testing Tasks

The following tasks must be completed:

1. Test Plan is prepared
2. Items to be tested are identified
3. Identify method of conducting tests
4. Assign personnel to conduct each test case
5. Conduct testing
6. Fix bugs and errors which are discovered
7. Create defect logs
8. Create test report

12 – Environmental Needs

For the testing of Wanderers, the following requirements and specifications must be met:

- Linux, Mac OS X, or Windows operating system
- Git (used for source version control)
- An appropriate web browser supporting JavaScript, preferably Google Chrome
- NPM installed on the user's machine (sometimes yarn)

13 – Staff and Training Needs

The Wanderers project requires a skilled and coordinated team to ensure the successful implementation and maintenance of the application.

Staff needs: Front-end and back-end developers, QA engineers, a project manager, and a release manager are needed. The development team will be responsible for implementing features, performing unit testing, and resolving issues. The QA team will conduct tests to

ensure application reliability and performance. The release manager will manage the release process, while the project manager will handle overall management of the project.

Training Needs: Team members need to be proficient in the core technologies used in the Wanderers project (Frontend: Next.js; Backend: Express.js; Database management: Supabase). Developers and QA engineers will receive training on CI/CD practices and automated testing tools. It’s essential for all team members to complete training in data privacy and security protocols to meet relevant standards. There will also be ongoing training sessions on using GitHub for version control and project management. The QA team will get training on the testing framework and reporting tools to maintain consistent quality assurance practices. All training materials and sessions will be documented and updated regularly.

14 – Responsibilities

Role	Responsibilities
Project Manager	Oversees project progress, approves and executes the project plan, assigns tasks, communicates project status, and manages team productivity and morale.
Release Manager	Manages the release process, ensuring smooth deployment of updates and new features, and verifies software reliability before release.
QA Manager	Implements and maintains testing strategies, conducts black-box and white-box testing, ensures test coverage, and oversees QA activities.
QA Engineer	Executes test plans, identifies bugs, and collaborates with developers to optimize system performance.
Frontend Developer	Develops the user interface, integrates front-end with back-end components, and performs usability testing.
Backend Developer	Develops back-end logic, integrates APIs and databases, and ensures system functionality for trip coordination and expense tracking.

15 – Schedule

The testing phase for Wanderers includes coding & unit testing, as well as integration & system testing. It will last for up to one month, with the timeline specified in the Project Plan. The personnel required for each process and the key person in charge for each process are detailed in the Project Plan as well. Coordination of the personnel required for each task, test team, development team, management and customer will be handled by the project manager. The planned activities are as follows:

- The number of days assigned to each test case will depend on their risk level. High risk will get 5 days, medium risk will get 4 days, and low risk will get 3 days. This is so that not too much time is wasted on testing low risk items.
- A daily backlog will be maintained to keep track of items which are yet to be completed. The QA Manager will continuously monitor this backlog and will help the QA Engineers if the backlog gets too large.
- For any bugs that are discovered, the result will be immediately updated on the JIRA board and classified as an issue and the Lead Developer will be notified of the same immediately.
- A one-week buffer is taken in the plan in case there aren't enough users for User Acceptance Testing.

16 – Planning Risks and Contingencies

The risks and methods to mitigate them are as follows:

Risk	Contingency
Shortage of testers	Members of the development team as well as management will pitch in to help with the testing if the QA team gets overwhelmed
Inadequate Training for Testers	The QA Manager and Lead Developer will ensure comprehensive training is provided to all testers to equip them with the necessary skills and knowledge for effective testing.
Improper communication amongst testers and developers	The daily testing backlog will ensure testing is on track. Testers will be required to report any bugs they discover immediately to the Lead Developer

17 – Approvals

For each testing type:

Role - Name	Test Type	Approval Criteria
Project Manager – Kim Seojin	Overall	The application works as required and user acceptance testing meets client needs before performing the official release for public usage
Lead Developer – Albert	White Box	System performs in accordance with functional requirements
Release & QA Manager - Isaac	Black Box	User Acceptance Testing meets acceptance criteria
Release & QA Engineer - Raghav	Black Box	All test cases are covered

18 – References

This master test plan is aligned with the following documents:

1. Wanderers Software Requirements Specification (SRS)
2. Wanderers Project Plan
3. Wanderers Quality Plan
4. Software Configuration Management Plan